

PROJECT PROPOSAL

**Development of an Integrated Student
Portal for Cavite State University —
Cavite City Campus**

PREPARED BY:

Cabidog, Erica Vera A.
Molina, Hazel Shanley B.
Palles, Romebhel M.
Terante, Ginalyn P.

BS Computer Science - 3A
College of Information and Communications Technology
Cavite State University — Cavite City Campus

SUBMITTED TO:

Mr. Armin Aragoncillo

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION

- Background of the Study 3
- Statement of the Problem 3
- Project Goals and Important 3
- Target User 4
- Scope and Limitations 4
- Definition of Terms 4

CHAPTER 2: REVIEW OF RELATED LITERATURE

- Academic Information Systems and User Awareness 6
- Security Management and Alphanumeric Authentication 6
- Local Perspectives on Digital Transformation..... 6
- Cloud Resilience and System..... 7
- Synthesis.....7

CHAPTER 3: METHODOLOGY

- Development Framework (Prototyping Model) 8
- System Architecture..... 8
- Operational Logic for Key Features 8
- Hardware and Software Requirements 9
- Testing and Data Security 9

REFERENCES 7

Chapter 1: Introduction

Background of the Project

Cavite State University (CvSU) – Cavite City Campus provides quality education to about 2,500 students. As the student population grows, the school needs a better way to handle digital data and communication. Right now, the existing student portal faces significant lag during busy times like enrollment and has a major security flaw: it uses default passwords that students cannot change. This means anyone who knows a student's ID can easily log into their account, putting private academic information and sensitive data at risk.

To fix these issues, we are developing a **Web-Based Integrated Student Portal** designed to be secure and easy to use. For the students, we added a **Settings** page where they can finally change their passwords using a 4-character alphanumeric format to meet modern privacy standards. We also included a **Notification Icon** that displays a red dot whenever a new update is posted, ensuring students stay informed about important school activities instantly. By using a web-based approach with cloud storage, this portal ensures that data stays protected and the system remains fast, providing a reliable digital space for the entire student community.

Statement of the Problem

The portal's performance is driven by a straightforward logic designed to make school tasks more secure and efficient. To fix the current security issues, the Password Management system requires students to change their default ID passwords to a personalized 4-character code using a mix of letters, numbers, and symbols, which is instantly updated in the secure database. Communication is handled by a Real-Time Notification system; the moment a new update or event is posted, a background listener detects the change and triggers a red dot on the student's notification icon as a clear visual cue. This alert is programmed to disappear automatically once the user views the message, ensuring the interface stays clean and updated. To bridge the support gap, the Ticketing Flow allows students to submit technical problems directly through the portal, creating a digital record that ensures concerns are documented and tracked. This setup ensures that every action—from securing an account to receiving an announcement—happens instantly and keeps the student connected to the campus.

Objects of the Study

The main goal of this project is to build a web-based portal that makes the campus more secure and easier for students to use. A key part of this is improving data privacy by requiring students to change their default passwords to a secure, 4-character version using a mix of letters, numbers, and symbols. Beyond security, the system aims to keep students informed through a real-time notification system. A red dot will appear on the notification icon whenever there is a new update, ensuring that students get the information they need immediately without having to check unofficial social media pages.

Additionally, the project focuses on making technical support much faster and more accessible. We are building a ticketing system so students can report portal issues directly through the system, creating a digital record that ensures their concerns are tracked and addressed. By moving everything to a reliable web-based platform, the portal will provide students with a clear, organized, and secure way to manage their university records and stay connected with campus updates.

Target Users

The system architecture is designed to prioritize the **Student** as the primary stakeholder to ensure a highly usable and secure operational framework. The development emphasizes the student interface—focusing on intuitive navigation, secure authentication, and efficient traffic management during peak periods. By centering the architecture on the student experience, the researchers ensure that the portal is a functional ecosystem where essential academic information such as grades, schedules, and enrollment status are delivered in real time. This approach guarantees that the portal serves as a reliable personal dashboard, keeping student data protected while maintaining a seamless connection to campus information.

Scope and Limitations

The scope of this enhancement includes secure user authentication, a real-time notification system, and a help desk ticketing module. The system is designed to provide quick access to schedules and grades while ensuring that students can manage their own privacy through a dedicated settings menu. To ensure a functional experience, the development focuses on the student as the primary stakeholder, prioritizing accessibility and ease of use within the interface. By requiring a 4-character alphanumeric password, the system ensures that personal data remains protected while remaining easy for the student to remember.

Beyond basic data management, the portal centralizes the student experience by allowing for the tracking of enrollment status and the viewing of official campus announcements. This focus creates a synchronized workflow where student inquiries and institutional updates are accessible within a single, unified platform through the use of a red dot alert on the notification icon. Although the system handles extensive academic information management, it remains limited to data viewing and support; it does not process financial payments due to the university's free tuition policies. Centering the development on the student ensures the portal's reliability and confirms that critical features, such as secure authentication and the automated notification system, function seamlessly for the end-user.

Definition of Terms

Alphanumeric. A mix of both letters (A-Z) and numbers (0-9). In this project, it refers to the security requirement to make student passwords harder to guess.

Backend. The "brain" of the portal that lives on the server. It handles the heavy work like checking passwords and storing all the information that users see on their screens.

Cloud Firestore. A digital storage space provided by Google, where the portal keeps

Firebase. The web-based platform is used to run the portal's database and security. It allows the system to update information instantly without needing a physical server on campus.

Information Loop. The constant flow of data between the Admin and the Student. For example, when an Admin posts a date, the student instantly sees a red notification bell.

Input Validation. A safety check that makes sure the information entered by a user is correct—like ensuring a new password is exactly four characters long—before the system saves it.

Real-Time Listener. A smart feature in the code that "watches" the database for changes. It is what makes the notification bell turn red the very second a new announcement is posted.

Ticketing System. A digital help desk where students can submit a report about a technical problem. It creates a trackable record so administrators can see the issue and fix it.

User Interface (UI). Everything the user sees and clicks on, such as the buttons, the calendar, the login screen, and the notification bell.

Chapter 2: Review of Related Literature

Academic Information Systems and User Awareness

The integration of digital portals in higher education has fundamentally changed how students interact with academic data. However, research by **Alatawi et al. (2020)** in their study on university portal usage suggests that many systems fail to reach their full potential due to a "lack of awareness" and "insufficient user guidance." Their findings indicate that students often miss critical deadlines when portals lack proactive communication tools. To address this, modern systems are now incorporating visual alert mechanisms, such as **notification dot bells** and **real-time alerts**, to bridge the gap between the institution and the student body. By utilizing color-coded signals—such as a bell icon that turns red—portals can effectively reduce the cognitive load on students while ensuring they remain informed of upcoming school events and administrative deadlines.

Security Management and Alphanumeric Authentication

A primary concern in the evolution of student portals is the transition from vulnerable default credentials to secure, user-managed authentication. **Reid and El-Najjar (2019)** argue that in academic environments, the "complexity" of a password—integrating uppercase letters, lowercase letters, numbers, and special characters—is a critical defense against unauthorized access. Their research supports the use of **alphanumeric logic** as a means to provide high security even within shorter character limits, which improves user memorability and reduces the frequency of locked accounts. This is particularly relevant when replacing static default passwords based on student ID numbers, as it empowers the user to take direct control over their academic privacy through a dedicated settings interface.

Local Perspectives on Digital Transformation

In the Philippine context, the shift from manual to digital processes is essential for institutional efficiency. **Aguinaldo (2017)**, in a study of a local college portal, highlights that centralized digital platforms significantly reduce the "manual coordination" required for viewing grades and schedules. This local research emphasizes that a student portal should serve as a "one-stop shop" that minimizes physical office visits. By providing a **Calendar of Events** and a structured **Help Desk Ticketing System**, institutions can automate support requests and information dissemination. This formalization of student inquiries ensures that technical complaints are tracked and resolved systematically, rather than through unorganized external messaging channels.

System Reliability and Performance

Modern web development focuses on making sure a system stays fast and reliable even when a lot of people are using it at the same time. By building this portal as a **Web-Based System** connected to **Firebase**, we ensure that the website can handle the "rush hour" periods on campus, such as during enrollment week. Since the database lives in the cloud, it can manage many students and staff members logging in simultaneously without the typical lag or system crashes found in older setups. This reliability is important because it ensures that critical tools, like the **Support Ticketing system** and the **Change Password module**, stay fast and responsive no matter how much traffic the portal receives.

Synthesis

The reviewed literature collectively underscores the necessity of moving toward secure, interactive, and cloud-resilient academic platforms. While earlier systems focused merely on data display, current research by Alatawi, Reid, and Aguinaldo proves that **security complexity**, **visual notifications**, and **structured support channels** are vital for user engagement and data integrity. This project synthesizes these findings by implementing a portal that utilizes a four-character alphanumeric security layer, a red-alert notification system, and a formalized ticketing desk. By adopting a decoupled cloud architecture, the system provides a reliable and future-proof solution tailored to the specific administrative and security needs of Cavite State University – Cavite City Campus.

References

- **Aguinaldo, R. V. (2017).** *Student Portal for St. Vincent College of Cabuyao*. [Unpublished Capstone Project]. St. Vincent College of Cabuyao.
- **Alatawi, S. S. T., Abdullah, N. S., & Miskon, S. (2020).** Research on student portal usage and its issues: A case study on Saudi Arabia universities, academic perspective. *International Journal of Engineering & Technology*, 9(1), 238-246.
- **National Privacy Commission. (2020).** *Advisory No. 2020-01: Data privacy and online learning*. Republic of the Philippines.
- **Reid, G., & El-Najjar, H. (2019).** An Analysis of Password Complexity and User Memorability in Academic Environments. *International Journal of Cyber Security and Digital Forensics (IJCSDF)*.
- **Shneiderman, B., et al. (2016).** *Designing the User Interface: Strategies for Effective Human-Computer Interaction* (6th ed.). Pearson.

Chapter 3: Methodology

Development Framework

The researchers adopted the Prototyping Model as the primary development framework. This iterative approach was selected to facilitate continuous testing and the modular addition of new features. Unlike traditional linear models, prototyping allowed the researchers to build functional versions of the Settings and Notification modules early in the process. This ensured that the specific 4-character alphanumeric validation and the real-time "listener" for the notification bell were refined through multiple iterations before the final deployment, ensuring high system reliability and user satisfaction.

System Architecture

The portal is built as a Web-Based System that connects a user-friendly frontend to a centralized Firebase backend. By using HTML, CSS, and JavaScript for the interface and Google Cloud Firestore for data storage, the system is designed to be accessible from any web browser without needing a complex local setup. This web-based approach solves the lag issues found in older systems by letting the cloud handle the heavy lifting. Instead of relying on a single, limited campus server that might crash during busy times like enrollment, this setup ensures that data flows smoothly to the student portal, keeping the system fast and responsive even when many people are logged in at once.

Operational Logic for Key Features

The portal's performance is driven by a straightforward logic designed to make school tasks more secure and efficient. To fix the current security issues, the Password Management system requires students to change their default ID passwords to a personalized 4-character code using a mix of letters, numbers, and symbols, which is instantly updated in the secure database. Communication is handled by a Real-Time Notification system; the moment a new update or event is posted, a background listener detects the change and triggers a red dot on the student's notification icon as a clear visual cue. This alert is programmed to disappear automatically once the user views the message, ensuring the interface stays clean and updated. To bridge the support gap, the Ticketing Flow allows students to submit technical problems directly through the

portal, creating a digital record that ensures concerns are documented and tracked. This setup ensures that every action—from securing an account to receiving an announcement—happens instantly and keeps the student connected to the campus.

Hardware and Software Requirements

The portal was built using a set of reliable tools to make sure the system stays stable and easy to use. For software, Visual Studio Code was used as the main workspace for writing and testing the code, while the Firebase Console acted as the system's "brain" for storing data and managing secure student logins. The portal is designed to work smoothly on any modern web browser like Chrome or Edge so that features like the red dot notification bell and the settings page function perfectly for every student.

On the hardware side, the project was developed using a laptop with enough processing power to handle local coding and live cloud updates at the same time. A stable internet connection was also necessary to ensure that the "real-time" features—like the instant ticketing and event alerts—could sync properly between the user's device and the database without any delays.

Testing and Data Security

To follow the **Data Privacy Act (RA 10173)**, the system was tested to make sure all information stays safe and accurate. **Validation Testing** was used on the password page to confirm that it correctly blocks any attempt to use a password that doesn't meet the 4-character secure format. Because this is a **web-based system** using **Firebase**, sensitive student data is stored securely in the cloud rather than being left exposed on the student's side of the website. This setup ensures that the portal provides a safe and private digital space for everyone at **CvSU – Cavite City Campus**.